

Unified Motion/Force/Impedance Control for Manipulators in Unknown Contact Environments Based on Robust Model-Reaching Approach

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Abstract—With the developments of intelligent and autonomous robotic technology, robots are usually designed to confront sophisticated tasks such as automated assembly, which requires both high-speed positioning capabilities and compliance with unknown contact environments. As we know, a high-performance motion tracking control in free space can achieve efficient and accurate positioning, while impedance or force control shows superior performance in terms of sensitive force and compliance with unknown contact environments. However, it is still challenging to achieve both high-performance motion tracking and compliance within one single control framework, especially in unknown contact environments. To this end, in this article, a unified motion/force/impedance approach for unknown contact environments is proposed by robust model-reaching control with dynamic trajectory adaptation. Specifically, the overall control scheme includes two loops: the outer loop replans the trajectories of motion and force in real time to meet the environmental constraints, which are estimated by the recursive least squares estimation law; in the inner loop, the robust model-reaching control law is proposed to realize a target model, which is designed to establish a dynamic relationship between the motion and force tracking errors of the replanned trajectories. Then, by changing the matrices of the target model, the compromise between motion tracking and force tracking can be achieved, as well as different control objectives. Experiments are conducted on a seven-degree-of-freedom manipulator to validate the advantages of the proposed scheme.

Index Terms—Force control, impedance control, manipulator control, trajectory adaptation, unknown environment.

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I. INTRODUCTION

IN THE past few decades, motivated by the automation of industries, motion trajectory tracking control techniques have been widely developed for mechatronic systems such as motors [1], vehicles [2], hydraulics [3]–[6], and especially the robotic systems [7]–[9]. In this case, manipulators are always designed to have higher stiffness in the mechanical structure as well, which can indirectly increase the bandwidth of the motion control. However, different from the “stiffer is better” rule in motion tracking, increasing sets of tasks nowadays need to deal with contact-sensitive objects and automated assembly, which require both high-speed positioning capabilities and compliance with unknown contact environments. Therefore, in addition to the motion control, the robotic systems nowadays give a strong emphasis on the compliance to the unknown contact environments as well.

Generally, the compliance of the robot system can be realized through two kinds of methods. One is by intentionally introducing compliant mechanical elements into the transmission train, such as the cable-driven manipulators [10], [11], the series elastic actuator [12] driven exoskeletons, and so on. However, the compliant elements in the mechanical structure decrease the control performance and stifle the possibility of high-precision motion control [13]. The other is through active compliance control algorithms, such as impedance [14] or force control. Robotic systems with impedance control are governed to be compliant to the unknown contact environments, but they have poor motion tracking performance in free space due to model uncertainties [15]. In addition, further research develops the position-based impedance control (i.e., admittance control), which employs the inner motion control loop to provide motion tracking in free space. However, it is easy to result in instability when the contact environment is comparatively stiff since the inner motion loop cannot track too aggressive trajectory [16]. As such, these approaches, either via impedance/force/admittance control or by introducing compliant elements into drive train, cannot guarantee both high-performance motion tracking in free space and compliance with unknown contact environments.

As we know, in many tasks such as automated assembly, a high-performance motion tracking control in free space can achieve efficient and accurate positioning [17], while impedance or force control shows superior performance in terms of sensitive force and compliance with unknown contact environments.

However, the aforementioned facts make it a prospective and challenging research topic to achieve both objectives within one single control framework.

Practically, for a unified control framework, many works designed the motion control, force control, impedance control, and hybrid motion/force control [18], respectively, and focused on the switching strategies. Ott *et al.* [15] proposed a unified impedance and admittance hybrid control scheme to continuously switch and interpolate between impedance and admittance control, which is effective in the known environments. On the basis of [15], Cavenago *et al.* [19] developed this approach to face unknown environment via a neural network, and Mei *et al.* [20] further combined the recursive least squares environment estimate law to face time-varying environment. However, these approaches have done an innovative work while do not include force tracking control, and there is a lack of verification experiments in practice.

From another perspective, the unified framework of motion/force/impedance control can be regarded as a tradeoff between the motion tracking and force regulation problem. Matinfar and Hashtrudi-Zaad [21] formulated the motion tracking and force regulation problem as an optimization problem, which can adjust the target impedance of robots for a wide range of environments. However, the linear-quadratic regulation was solved offline and based on the known environment dynamics in this study. Ge *et al.* [22], [23] improved the approach in [21] by offline solving the adaptive dynamic programming with the unknown environment's dynamics. These effective results were obtained under the assumption that the inner motion control loop could track the optimized reference trajectory well. However, the inner loop cannot track too aggressive trajectory for the limited bandwidth. Different from the motion-based approach, Yao *et al.* [24] proposed a generalized impedance model, which directly establishes the dynamic relationship between the motion tracking error and the force tracking error in the control law, so that the motion control, impedance control, and hybrid motion/force control can be realized just by simply changing the matrices in the target model. It is widely used in recent studies, such as the impedance control of [25]. However, the transition between different objectives was not considered in this method, and it not only required both motion and force reference trajectories in advance but also required these trajectories to conform to unknown environment dynamics, so the environment dynamics was assumed to be known in advance.

The above literature review shows that the challenges of developing a unified motion/force/impedance control in unknown environment are twofold: first, it is difficult to formulate a unified representation of different controllers; the second is that the process uncertainties in unknown contact environments make it difficult to accurately switch between different control objectives. Faced with the aforementioned challenges, this article further develops the general model proposed by Yao *et al.* [24] and realizes the unified control of manipulator motion/force/impedance in unknown environment through the combination of robust model-reaching control law and dynamic trajectory adaptation. The novelties and contributions of this article include the following.

- 1) In order to replan the desired trajectory to conform to the unknown contact environment, the dynamic trajectory adaptation approach is proposed based on the real-time estimated environment dynamics and the kinematic constraints.
- 2) The generalized model and robust model-reaching control are utilized to formulate a unified representation of different controllers based on the replanned trajectories. The control performance and stability are guaranteed and proved even during the variation of the target model.
- 3) A unified framework of motion/force/impedance control in unknown contact environments is developed by integrating these two. Unlike the traditional position/force switching control, the proposed approach not only allows the accuracy of motion tracking and the contact compliance to be effectively combined, but also enables a smooth transition between these two, especially in the unknown contact environments.

The superiority of the scheme is illustrated by comparative experiments on the Franka Emika Panda seven-degree-of-freedom (DOF) manipulators; results show the free-space motion tracking accuracy and contact compliance. Furthermore, the two-DOF continuous motion and force tracking performance are validated by another experiment.

The rest of this article is organized as follows. In Section II, the problem is formulated with the dynamics of the manipulator and the environment. The unified motion/force/impedance control in unknown environment is achieved in Section III. In Section IV, the proposed approach is verified experimentally on a seven-DOF manipulator. Finally, Section V concludes this article.

II. PROBLEM FORMULATION

In this section, the system under study includes a general n -DOF rigid manipulator dynamics and an unknown environment dynamics, and the problem of unified motion/force/impedance control for manipulators in unknown contact environments will be stated based on these two dynamics.

A. Manipulator Dynamics

First, the dynamics of an n -DOF manipulator in joint space can be written as

$$\begin{aligned} M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) + \tau_f + \tilde{f}(t) \\ = \tau - J^T(q)F_{\text{ext}}(t) \end{aligned} \quad (1)$$

where $q \in R^{n \times 1}$ denotes the joint position of the manipulator, and $\dot{q}$ and $\ddot{q}$ denote the velocity and acceleration of the robot joints, respectively, $M(q) \in R^{n \times n}$ is the inertia matrix, $C(q, \dot{q})\dot{q} \in R^{n \times 1}$ denotes the Coriolis and centrifugal force vector, $G(q) \in R^{n \times 1}$ represents the gravitational term, τ_f is the vector of friction torques, $\tilde{f}(t) \in R^{n \times 1}$ represents the modeling error and external disturbance, $\tau \in R^{n \times 1}$ is the vector of applied joint torque, $F_{\text{ext}}(t) \in R^{6 \times 1}$ is the vector of external force/moment between the end-effector and the unknown environment, and $J^T(q) \in R^{n \times 6}$ represents the transpose of the geometric Jacobian matrix.

As the unified control is designed in Cartesian coordinate, consider the robot kinematics given by

$$\mathbf{x}(t) = \phi(\mathbf{q}) \quad (2)$$

$$\dot{\mathbf{x}}(t) = \mathbf{J}(\mathbf{q})\dot{\mathbf{q}} \quad (3)$$

where $\mathbf{x}(t) \in \mathbb{R}^{6 \times 1}$ represents the position and orientation of the end-effector in the Cartesian space and $\dot{\mathbf{x}}(t)$ denotes the translation and angular velocity of the end-effector.

Remark 1: It should be noted that $\mathbf{x}(t)$ and $\dot{\mathbf{x}}(t)$ do not have the time derivative relations here, since the angular velocity $\dot{\mathbf{x}}(t)$ is described based on the body frame, which is directly calculated from the geometric Jacobian matrix $\mathbf{J}(\mathbf{q})$ and the joint velocity $\dot{\mathbf{q}}$; refer to [26] for detail.

Differentiating (3) with respect to time results in

$$\ddot{\mathbf{x}}(t) = \dot{\mathbf{J}}(\mathbf{q})\dot{\mathbf{q}} + \mathbf{J}(\mathbf{q})\ddot{\mathbf{q}} \quad (4)$$

where $\ddot{\mathbf{x}}$ represents the translation and angular acceleration of the end-effector.

Then, substituting the kinematics (2)–(4) into the joint space dynamics (1), the dynamic model of the manipulator can be derived as follows in operational space:

$$\mathbf{M}_O(\mathbf{q})\ddot{\mathbf{x}} + \mathbf{C}_O(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{x}} + \mathbf{G}_O(\mathbf{q}) + \mathbf{F}_f + \tilde{\mathbf{F}} = \mathbf{T} - \mathbf{F}_{\text{ext}} \quad (5)$$

$$\mathbf{M}_O(\mathbf{q}) = \mathbf{J}^{\dagger T} \mathbf{M} \mathbf{J}^{\dagger}$$

$$\mathbf{C}_O(\mathbf{q}, \dot{\mathbf{q}}) = \mathbf{J}^{\dagger T} (\mathbf{C} - \mathbf{M} \mathbf{J}^{\dagger} \dot{\mathbf{J}}) \mathbf{J}^{\dagger}$$

$$\mathbf{G}_O(\mathbf{q}) = \mathbf{J}^{\dagger T} \mathbf{G}, \quad \mathbf{F}_f = \mathbf{J}^{\dagger T} \boldsymbol{\tau}_f$$

$$\tilde{\mathbf{F}} = \mathbf{J}^{\dagger T} \tilde{\mathbf{f}}, \quad \mathbf{T} = \mathbf{J}^{\dagger T} \boldsymbol{\tau} \quad (6)$$

where $\mathbf{J}^{\dagger}$ denotes the pseudoinverse Jacobian matrix.

Property 1: $\mathbf{M}_O(\mathbf{q})$ is symmetric positive-definite (s.p.d.) matrix, with $k'_r \mathbf{I}_{n \times n} \leq \mathbf{M}_O \leq k''_r \mathbf{I}_{n \times n}$, where k'_r and k''_r denote positive scalars.

Property 2: $\dot{\mathbf{M}}_O(\mathbf{q}) - 2\mathbf{C}_O(\mathbf{q}, \dot{\mathbf{q}})$ is a skew-symmetric matrix, i.e., $\frac{1}{2} \boldsymbol{\zeta}^T \dot{\mathbf{M}}_O(\mathbf{q}) \boldsymbol{\zeta} = \boldsymbol{\zeta}^T \mathbf{C}_O(\mathbf{q}, \dot{\mathbf{q}}) \boldsymbol{\zeta} \forall \boldsymbol{\zeta} \in \mathbb{R}^n$.

Let $\Delta \bullet$ and $\hat{\bullet}$ denote the modeling error and estimated value of $\bullet$, respectively, i.e., $\Delta \bullet = \bullet - \hat{\bullet}$. It should be noted that the nominal value usually cannot be exactly known in the early identification experiments; only estimated values $\hat{\mathbf{M}}_O$, $\hat{\mathbf{C}}_O$, $\hat{\mathbf{G}}_O$, and $\hat{\mathbf{F}}_f$ are available in (5).

Assumption: The modeling error are bounded by known bounds, i.e.

$$\begin{aligned} \|\Delta \mathbf{M}_O(\mathbf{q})\| &= \|\mathbf{M}_O(\mathbf{q}) - \hat{\mathbf{M}}_O(\mathbf{q})\| \leq \delta \mathbf{M}_O \\ \|\Delta \mathbf{C}_O(\mathbf{q}, \dot{\mathbf{q}})\| &= \|\mathbf{C}_O(\mathbf{q}, \dot{\mathbf{q}}) - \hat{\mathbf{C}}_O(\mathbf{q}, \dot{\mathbf{q}})\| \leq \delta \mathbf{C}_O \\ \|\Delta \mathbf{G}_O(\mathbf{q})\| &= \|\mathbf{G}_O(\mathbf{q}) - \hat{\mathbf{G}}_O(\mathbf{q})\| \leq \delta \mathbf{G}_O \\ \|\Delta \mathbf{F}_f\| &= \|\mathbf{F}_f - \hat{\mathbf{F}}_f\| \leq \delta \mathbf{F}_f, \quad \|\tilde{\mathbf{F}}\| \leq \delta \tilde{\mathbf{F}} \end{aligned} \quad (7)$$

where $\|\bullet\|$ denotes a norm of $\bullet$, which is a vector or a matrix; without loss of generality, $\|\bullet\|_2$ is used, i.e., $\|\bullet\| = \sigma_{\max}(\bullet) = \sqrt{\lambda_{\max}(\bullet^T \bullet)}$, where $\sigma(\bullet)$ denotes a singular value of $\bullet$, $\lambda(\bullet)$ means the eigenvalue of $\bullet$. The positive bounds $\delta \mathbf{M}_O$, $\delta \mathbf{C}_O$, $\delta \mathbf{G}_O$, $\delta \mathbf{F}_f$, and $\delta \tilde{\mathbf{F}}$ are assumed known.

B. Unknown Environment Dynamics

In practice, manipulators can obtain approximate environmental information through external sensors (such as cameras), but before establishing contact, it is often impossible to fully understand the accurate contact point and contact stiffness. Under the hypothesis of a stable and single contact, the unknown environment dynamics in this article is formulated with unknown contact position and unknown contact stiffness. The most commonly used environment model is the Kelvin–Voigt linear model as in [27], where the environments can be modeled as a linear damper–spring system as follows:

$$\mathbf{K}_e(\mathbf{x} - \mathbf{x}_e) + \mathbf{D}_e \dot{\mathbf{x}} = \mathbf{F}_e \quad (8)$$

where $\mathbf{K}_e$ and $\mathbf{D}_e$ represent the stiffness and damping of the environments, respectively, $\mathbf{x}_e$ denotes the environment equilibrium position, and $\mathbf{F}_e$ is the contact force/moment between the environment and the manipulator, i.e., $\mathbf{F}_e = -\mathbf{F}_{\text{ext}}$.

Especially, when the stiffness $\mathbf{K}_e = 0$, and the damping $\mathbf{D}_e$ is small enough, the environment dynamics can be seen as free space. Hence, (8) can represent the unknown environment dynamics both for contact and noncontact situations.

C. Unified Motion/Force/Impedance Control Problem Statement

The problem of unified motion/force/impedance control for manipulators in unknown environment dynamics is stated based on (5) and (8) in this subsection. Considering the desired motion and force trajectories $\mathbf{x}_d(t)$ and $\mathbf{F}_d(t)$ in the Cartesian space, the motion and force tracking errors can be represented as $\mathbf{e}_p = \mathbf{x}(t) - \mathbf{x}_d(t)$ and $\mathbf{e}_f = \mathbf{F}_{\text{ext}}(t) - \mathbf{F}_d(t)$, respectively. The unified motion/force/impedance control of manipulators can be stated as that of determining a control law so that the motion and force errors $\mathbf{e}_p$ and $\mathbf{e}_f$ achieve the following target model:

$$\mathbf{M}_d \ddot{\mathbf{e}}_p + \mathbf{D}_d \dot{\mathbf{e}}_p + \mathbf{K}_d \mathbf{e}_p = -\mathbf{K}_f \mathbf{e}_f \quad (9)$$

where $\mathbf{M}_d$, $\mathbf{D}_d$, $\mathbf{K}_d$, $\mathbf{K}_f \in \mathbb{R}^{n \times n}$ are desired matrices, which can be changed with respect to control objectives, and usually chosen as diagonal matrices to get decoupled responses in the Cartesian space.

Remark 2: It is important that the desired trajectories $\mathbf{x}_d(t)$ and $\mathbf{F}_d(t)$ in the target model (9) should conform to the environment dynamics (8), and $\mathbf{M}_d$, $\mathbf{D}_d$, $\mathbf{K}_d$, and $\mathbf{K}_f$ should be chosen properly when interacting with environments.

Remark 3: The target model (9) is actually a tradeoff between motion tracking and force tracking of manipulators. The motion tracking is emphasized by assigning $\mathbf{K}_d \gg \mathbf{K}_f$; then, the steady-state motion tracking error $\mathbf{e}_p$ is relatively small, whereas the force tracking error $\mathbf{e}_f$ is large. Otherwise, force tracking can be achieved with $\mathbf{K}_d \ll \mathbf{K}_f$ as well.

Specifically, diversity control objectives can be realized for manipulators via the target model (9) if the desired motion/force trajectory $\mathbf{x}_d(t)/\mathbf{F}_d(t)$ and the matrices $\mathbf{M}_d$, $\mathbf{D}_d$, $\mathbf{K}_d$, and $\mathbf{K}_f$ are well settled, such as the motion control, the impedance control, the force control, the hybrid motion/force control, and so on [24].

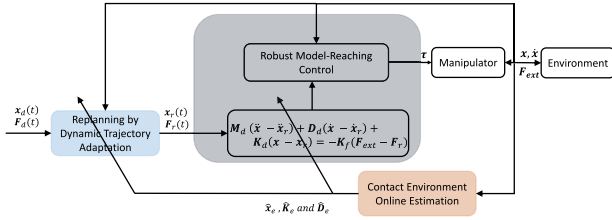


Fig. 1. Overall control scheme of the proposed approach.

As such, to achieve the unified motion/force/impedance control in unknown contact environments, first, the motion and force tracking performance of the manipulator should be regulated by adjusting the matrices M_d , D_d , K_d , and K_f in the target model (9). Meanwhile, the desired motion and force trajectories in the target model are also required to meet the geometric and dynamic constraints of the unknown contact environments (8). In the next section, the robust model-reaching-based approach is proposed for these objectives.

III. ROBUST MODEL-REACHING CONTROL WITH DYNAMIC TRAJECTORY ADAPTATION

The overall control architecture of the closed-loop system is given in Fig. 1; it includes two loops: in the outer loop, both the desired motion and force trajectories $x_d(t)$ and $F_d(t)$ are replanned according to the environmental constraints, which are estimated by the environment estimation algorithm. Meanwhile, with the replanned trajectories $x_r(t)$ and $F_r(t)$, the target model in the inner loop is also changed according to the control objectives and estimated environment dynamics. Finally, the robust model-reaching law is designed to achieve the target model under the bounded uncertainties (7), thereby realizing the compromise between motion tracking and force tracking.

A. Environment Online Estimation

The unknown environment dynamics in (8) is identified online via recursive least squares estimation (RLSE), including the environment equilibrium position x_e , stiffness K_e , and damping D_e . Assuming that the environment dynamics is fully decoupled, the environment dynamics of each environment dimension can be written as the regressor form

$$F_{e,i} = \varphi_i^T \theta_i \quad (10)$$

where $\bullet_i$, $i = 1, 2, \dots, 6$, denote the i th DOF, and φ_i and θ_i are defined as

$$\varphi_i = [x_i - \hat{x}_{e,i}, \dot{x}_i]^T, \quad \theta_i = [K_{e,i}, D_{e,i}]^T. \quad (11)$$

The RLSE method is used to find an estimated $\hat{\theta}_i$ of each DOF, which can minimize the performance index function

$$J(t, \hat{\theta}_i) = \int_0^t \|W(t, \tau)[F_{e,i}(\tau) - \varphi_i(\tau)^T \hat{\theta}_i]\|^2 d\tau \quad (12)$$

where $W(t, \tau)$ represents the weighting matrix.

Then, the RLSE update equation can be derived as follows:

$$\begin{aligned} \hat{\theta}_i(t) &= \Gamma_i(t) \varphi_i(t) \epsilon_i(t), \quad \epsilon_i(t) = F_{e,i} - \varphi_i^T \hat{\theta}_i \\ D_{\Gamma,i} &= \mu_1(t) \Gamma_i(t) - \mu_2(t) \Gamma_i(t) \varphi_i(t) \varphi_i^T \Gamma_i(t) \\ \dot{\Gamma}_i(t) &= \begin{cases} D_{\Gamma,i}, & \text{if } \lambda_{\max}(\Gamma_i(t)) < \rho_M \\ 0, & \text{otherwise} \end{cases} \end{aligned} \quad (13)$$

where $\epsilon_i(t)$ represents the prediction error; the covariance is initialized to a value bounded as $0 < \Gamma_i(0) < \rho_M I_p$; and $\mu_1(t)$ and $\mu_2(t)$ are the weighting, which are chosen as $0 < \mu_1(t) \leq 1$ and $0 < \mu_2(t) \leq 2$ for stability.

Theorem 1: The parameter estimation error ϵ_i is uniformly bounded with the RLSE law (13), i.e., $\hat{\theta}_i(t) \in L_\infty$.

Proof 1: Denote $\epsilon_i(t) = -\varphi_i^T \tilde{\theta}_i$, and define the positive function $V_{\theta,i} = \tilde{\theta}_i(t)^T \Gamma_i(t)^{-1} \tilde{\theta}_i(t)$; then, differentiating $V_{\theta,i}$ gives

$$\begin{aligned} \dot{V}_{\theta,i}(t) &= 2\tilde{\theta}_i(t)^T \Gamma_i(t)^{-1} \dot{\tilde{\theta}}_i(t) + \tilde{\theta}_i(t)^T \frac{d\Gamma_i(t)^{-1}}{dt} \tilde{\theta}_i(t) \\ &= 2\tilde{\theta}_i(t)^T \varphi_i(t) \epsilon_i(t) + \tilde{\theta}_i(t)^T [-\mu_1(t) \Gamma_i(t)^{-1} \\ &\quad + \mu_2(t) \varphi_i(t) \varphi_i^T \Gamma_i(t)^{-1}] \tilde{\theta}_i(t) \\ &= -2\epsilon_i(t)^2 - \mu_1(t) \tilde{\theta}_i(t)^T \Gamma_i(t)^{-1} \tilde{\theta}_i(t) + \mu_2(t) \epsilon_i(t)^2. \end{aligned} \quad (14)$$

Remark 4: The estimated value $\hat{x}_{e,i}$ in φ_i is obtained when the manipulator first contacts with environments.

Remark 5: The environment damping usually cannot be estimated with sufficient accuracy during the first contact establishment process since the environment dynamics is not excited sufficiently, i.e., persistent excitation condition is not always satisfied. Continuous contact and variations of the contact force should be exerted on the environment for accurate estimations.

B. Dynamic Trajectory Adaptation

In actual applications, before the environment estimation, the environment geometric and dynamic characteristics are often not sufficiently known, so that it is hard to give the desired motion trajectory $x_d(t)$ and the force trajectory $F_d(t)$, which meet the environment well. As such, the desired motion trajectory $x_d(t)$ and the force trajectory $F_d(t)$ are replanned based on the environmental constraints estimated in previous subsection. The objectives of the dynamic trajectory adaptation are mainly twofold: On the one hand, the online replanned trajectories $x_r(t)$ and $F_r(t)$ not only meet the environment characteristics and manipulator kinematic constraints, but also converge to the desired trajectories as much as possible. On the other hand, the initial value of the online replanned reference trajectories $x_r(t)$ should be calibrated to the real-time measurements $x(t)$ at the beginning of each replanning sampling period, i.e., the motion tracking error is small at the initial instant of each replanning sampling period, so the mismatches of the motion and velocity when switching the control objectives are eliminated.

The objectives of dynamic trajectory adaptation can be formulated as a constrained optimization problem via model-predictive control in each sampling period

$$\begin{aligned} \min_{\mathbf{x}_r, \mathbf{F}_r} \quad & J = \int_{t_0}^{t_0+t_p} \left[\frac{1}{2} (\mathbf{x}_r - \mathbf{x}_d)^T \mathbf{Q} (\mathbf{x}_r - \mathbf{x}_d) \right. \\ & \left. + \frac{1}{2} (\mathbf{F}_r - \mathbf{F}_d)^T \mathbf{R} (\mathbf{F}_r - \mathbf{F}_d) \right] dt \\ \text{s.t.} \quad & \begin{cases} \hat{\mathbf{K}}_e(\mathbf{x}_r - \hat{\mathbf{x}}_e) + \hat{\mathbf{D}}_e \dot{\mathbf{x}}_r = -\mathbf{F}_r \\ \mathbf{x}_r(t_0) = \mathbf{x}(t_0) \\ \dot{\mathbf{x}}_{\min} \leq \dot{\mathbf{x}}_r \leq \dot{\mathbf{x}}_{\max} \\ \ddot{\mathbf{x}}_{\min} \leq \ddot{\mathbf{x}}_r \leq \ddot{\mathbf{x}}_{\max} \\ \mathbf{F}_{\min} \leq \mathbf{F}_r \leq \mathbf{F}_{\max} \end{cases} \end{aligned} \quad (15)$$

where t_0 and t_p represent the initial time and prediction horizon of each replanning sampling period, respectively; $\mathbf{Q}$ and $\mathbf{R}$ denote the weighting matrices of motion and force, respectively; $\dot{\mathbf{x}}_r$ and $\ddot{\mathbf{x}}_r$ represent the velocity and acceleration of the replanned motion trajectory, respectively; $\dot{\mathbf{x}}_{\min}$ and $\dot{\mathbf{x}}_{\max}$ are the minimum and maximum velocity constraints, respectively; $\ddot{\mathbf{x}}_{\min}$ and $\ddot{\mathbf{x}}_{\max}$ are the minimum and maximum acceleration constraints, respectively; also, $\mathbf{F}_{\min}$ and $\mathbf{F}_{\max}$ are the bound of replanned force.

The optimization problem in (15) is converted into a quadratic programming (QP) problem and solved real time with this form

$$\begin{aligned} \min \quad & \frac{1}{2} \mathbf{f}_{\text{seq}}^T \mathbf{H}(t_0) \mathbf{f}_{\text{seq}} + \mathbf{f}_{\text{seq}}^T \mathbf{g}(t_0) \\ \text{s.t.} \quad & \begin{cases} \mathbf{lb} \mathbf{A}(t_0) \leq \mathbf{A} \mathbf{f}_{\text{seq}} \leq \mathbf{ub} \mathbf{A}(t_0) \\ \mathbf{lb}(t_0) \leq \mathbf{f}_{\text{seq}} \leq \mathbf{ub}(t_0) \end{cases} \end{aligned} \quad (16)$$

where $\mathbf{f}_{\text{seq}}$ is a sequence of $\mathbf{F}_r$ on time; $\mathbf{H}(t_0)$ is the Hessian matrix related to $\mathbf{Q}$, $\mathbf{R}$, and the environmental constraints in (15); $\mathbf{g}(t_0)$ is the gradient vector concerning with the $\mathbf{x}(t_0)$, $\mathbf{Q}$, and environmental constraints; $\mathbf{lb} \mathbf{A}(t_0)$ and $\mathbf{ub} \mathbf{A}(t_0)$ are derived from the kinematic constraints of $\mathbf{x}_r$; and $\mathbf{lb}(t_0)$ and $\mathbf{ub}(t_0)$ are related to the bound of the replanned force.

With the replanned trajectories, note the motion tracking error and force tracking error as $\mathbf{e}_{pr} = \mathbf{x}(t) - \mathbf{x}_r(t)$ and $\mathbf{e}_{fr} = \mathbf{F}_{\text{ext}}(t) - \mathbf{F}_r(t)$, respectively. The target model in (9) can be derived as

$$\mathbf{M}_d(t) \ddot{\mathbf{e}}_{pr} + \mathbf{D}_d(t) \dot{\mathbf{e}}_{pr} + \mathbf{K}_d(t) \mathbf{e}_{pr} = -\mathbf{K}_f(t) \mathbf{e}_{fr} \quad (17)$$

where the error $\mathbf{e}_{pr}$ is small at the initial instant of each sampling period.

Remark 6: During the variation of the target model, the passivity should be guaranteed under constraints with respect to the derivatives of these matrices. Define the energy function as $V_e = \frac{1}{2} \dot{\mathbf{e}}_{pr}^T \mathbf{M}_d(t) \dot{\mathbf{e}}_{pr} + \frac{1}{2} \mathbf{e}_{fr}^T \mathbf{K}_d(t) \mathbf{e}_{fr}$, and its derivative is

$$\begin{aligned} \dot{V}_e = & \dot{\mathbf{e}}_{pr}^T (-\mathbf{K}_f(t) \mathbf{e}_{fr}) \\ & + \left(\frac{1}{2} \dot{\mathbf{e}}_{pr}^T (\dot{\mathbf{M}}_d(t) - 2\mathbf{D}_d(t)) \dot{\mathbf{e}}_{pr} + \frac{1}{2} \mathbf{e}_{fr}^T \dot{\mathbf{K}}_d(t) \mathbf{e}_{fr} \right) \end{aligned} \quad (18)$$

where the increase of inertia matrix and stiffness matrix may destroy the passivity on the pair $\langle \dot{\mathbf{e}}_{pr}, -\mathbf{K}_f \mathbf{e}_{fr} \rangle$; refer to [28] for details.

C. Robust Model-Reaching Control Law

The robust model-reaching strategy is designed to achieve the target model with replanned trajectories (17) in this subsection.

First, a dynamic compensator is formulated as

$$\dot{\mathbf{z}} = \mathbf{A} \mathbf{z} + \mathbf{K}_{pz}(t) \mathbf{e}_{pr} + \mathbf{K}_{dz}(t) \dot{\mathbf{e}}_{pr} + \mathbf{K}_{fz}(t) \mathbf{e}_{fr} \quad (19)$$

where $\mathbf{z}$ is the n -dimensional state vector of the compensator, $\mathbf{A} \in \mathbb{R}^{n \times n}$ is a diagonal matrix, which is also negative semidefinite, and $\mathbf{K}_{pz}(t)$, $\mathbf{K}_{dz}(t)$, and $\mathbf{K}_{fz}(t) \in \mathbb{R}^{n \times n}$ are matrices related to the target model.

With the compensator state $\mathbf{z}$, the switching function is formed as follows:

$$\mathbf{s}(\mathbf{e}_{pr}, \dot{\mathbf{e}}_{pr}, \mathbf{z}) = \dot{\mathbf{e}}_{pr} + \mathbf{F}_1 \mathbf{e}_{pr} + \mathbf{F}_2 \mathbf{z} \quad (20)$$

where $\mathbf{F}_1, \mathbf{F}_2 \in \mathbb{R}^{n \times n}$ are any diagonal constant matrices, and $\mathbf{F}_2$ should be chosen as a nonsingular matrix.

Differentiating (20) with respect to time as

$$\dot{\mathbf{s}} = \ddot{\mathbf{e}}_{pr} + \mathbf{F}_1 \dot{\mathbf{e}}_{pr} + \mathbf{F}_2 \dot{\mathbf{z}}. \quad (21)$$

Substituting the dynamic compensator (19) into (21) and noting (20), $\mathbf{z}$ can be eliminated

$$\begin{aligned} \ddot{\mathbf{e}}_{pr} + (\mathbf{F}_1 - \mathbf{F}_2 \mathbf{A} \mathbf{F}_2^{-1} + \mathbf{F}_2 \mathbf{K}_{dz}(t)) \dot{\mathbf{e}}_{pr} + (\mathbf{F}_2 \mathbf{K}_{pz}(t) \\ - \mathbf{F}_2 \mathbf{A} \mathbf{F}_2^{-1} \mathbf{F}_1) \mathbf{e}_{pr} = -\mathbf{F}_2 \mathbf{K}_{fz}(t) \mathbf{e}_{fr} + \dot{\mathbf{s}} - \mathbf{F}_2 \mathbf{A} \mathbf{F}_2^{-1} \mathbf{s}. \end{aligned} \quad (22)$$

Since the matrices $\mathbf{A}$ and $\mathbf{F}_2$ are diagonal matrices, the relationship $\mathbf{F}_2 \mathbf{A} \mathbf{F}_2^{-1} = \mathbf{A}$ can be derived. Choose the matrices $\mathbf{K}_{pz}(t)$, $\mathbf{K}_{dz}(t)$, and $\mathbf{K}_{fz}(t)$ as

$$\begin{aligned} \mathbf{K}_{dz}(t) &= \mathbf{F}_2^{-1} (\mathbf{M}_d(t)^{-1} \mathbf{D}_d(t) - \mathbf{F}_1 + \mathbf{A}) \\ \mathbf{K}_{pz}(t) &= \mathbf{F}_2^{-1} (\mathbf{M}_d(t)^{-1} \mathbf{K}_d(t) + \mathbf{A} \mathbf{F}_1) \\ \mathbf{K}_{fz}(t) &= \mathbf{F}_2^{-1} \mathbf{M}_d(t)^{-1} \mathbf{K}_f(t). \end{aligned} \quad (23)$$

Then, (24) can be derived by substituting (23) into (22)

$$\begin{aligned} \mathbf{M}_d(t) \ddot{\mathbf{e}}_{pr} + \mathbf{D}_d(t) \dot{\mathbf{e}}_{pr} + \mathbf{K}_d(t) \mathbf{e}_{pr} = -\mathbf{K}_f(t) \mathbf{e}_{fr} \\ + \mathbf{M}_d(t) (\dot{\mathbf{s}} - \mathbf{A} \mathbf{s}) \end{aligned} \quad (24)$$

where the target model (17) can be achieved in the sliding mode $\{\mathbf{s} \rightarrow 0, \dot{\mathbf{s}} \rightarrow 0\}$ [29].

As such, for the manipulator described by (5), the manipulator can achieve the desired target model under the bounded uncertainties if the following control input is applied:

$$\begin{aligned} \mathbf{T} = & \hat{\mathbf{M}}_O \ddot{\mathbf{x}}_{\text{eq}} + \hat{\mathbf{C}}_O \dot{\mathbf{x}}_{\text{eq}} + \hat{\mathbf{G}}_O \\ & + \hat{\mathbf{F}}_f + \mathbf{F}_{\text{ext}} - \mathbf{K}_s \mathbf{s} - d \frac{\mathbf{s}}{\|\mathbf{s}\|} \end{aligned} \quad (25)$$

where

$$\dot{\mathbf{x}}_{\text{eq}} = \dot{\mathbf{x}}_r(t) - \mathbf{F}_1 \mathbf{e}_{pr} - \mathbf{F}_2 \mathbf{z}, \quad \ddot{\mathbf{x}}_{\text{eq}} = \ddot{\mathbf{x}}_r(t) - \mathbf{F}_1 \dot{\mathbf{e}}_{pr} - \mathbf{F}_2 \dot{\mathbf{z}}. \quad (26)$$

$\mathbf{K}_s$ is a defined s.p.d. matrix and d is a positive scalar satisfying

$$d \geq \delta M_O \|\ddot{\mathbf{x}}_{\text{eq}}\| + \delta C_O \|\dot{\mathbf{x}}_{\text{eq}}\| + \delta G_O + \delta \tilde{F} + \varepsilon \quad (27)$$

where ε is a positive scalar.

Theorem 2: With the designed control input (25), the manipulator system described in (5) achieves the desired target model (17) under the bounded uncertainties (7).

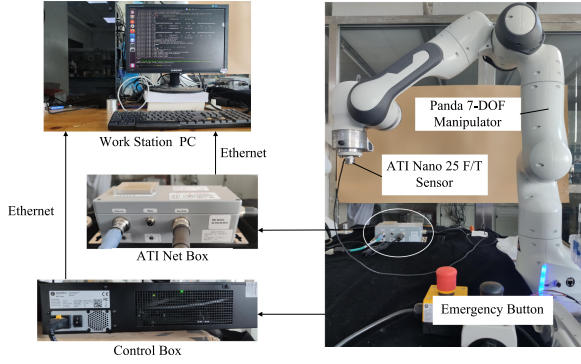


Fig. 2. Overview of the experiment system.

Proof 2: Define a positive-definite function as $V = \frac{1}{2} \mathbf{s}^T \mathbf{M}_O \mathbf{s}$. From Property 1, the following inequality can be derived:

$$\frac{1}{2} k_r' \|\mathbf{s}\|^2 \leq V \leq \frac{1}{2} k_r'' \|\mathbf{s}\|^2. \quad (28)$$

Differentiating V , using Property 2, and substituting the control input $\mathbf{T}$ (25) into $\dot{V}$, we obtain

$$\begin{aligned} \dot{V} &= \mathbf{s}^T \mathbf{M}_O \dot{\mathbf{s}} + \frac{1}{2} \mathbf{s}^T \dot{\mathbf{M}}_O \mathbf{s} = \mathbf{s}^T (\mathbf{M}_O \dot{\mathbf{s}} + \mathbf{C}_O \mathbf{s}) \\ &= -\mathbf{s}^T \mathbf{K}_s \mathbf{s} - d \|\mathbf{s}\| + \mathbf{s}^T [-\Delta \mathbf{M}_O \dot{\mathbf{x}}_{eq} - \Delta \mathbf{C}_O \dot{\mathbf{x}}_{eq} \\ &\quad - \Delta \mathbf{G}_O - \Delta \mathbf{F} - \Delta \tilde{\mathbf{F}}] \\ &\leq -\mathbf{s}^T \mathbf{K}_s \mathbf{s} - d \|\mathbf{s}\| + \mathbf{s}^T [\|\Delta \mathbf{M}_O\| \|\dot{\mathbf{x}}_{eq}\| \\ &\quad + \|\Delta \mathbf{C}_O\| \|\dot{\mathbf{x}}_{eq}\| + \|\Delta \mathbf{G}_O\| + \|\Delta \mathbf{F}\| + \|\Delta \tilde{\mathbf{F}}\|] \\ &\leq -\lambda_{\min}(\mathbf{K}_s) \|\mathbf{s}\|^2 - \varepsilon \|\mathbf{s}\| \\ &\leq -\frac{2\lambda_{\min}(\mathbf{K}_s)}{k_r''} V - \varepsilon \sqrt{\frac{2}{k_r''}} \sqrt{V}. \end{aligned} \quad (29)$$

The actual input in joint space can be derived from (6) and (25) as follows:

$$\begin{aligned} \boldsymbol{\tau} &= \hat{\mathbf{M}} \ddot{\mathbf{q}}_{eq} + \hat{\mathbf{C}} \dot{\mathbf{q}}_{eq} + \hat{\mathbf{G}} + \hat{\boldsymbol{\tau}}_f + \mathbf{J}^T \mathbf{F}_{ext} - \mathbf{J}^T \\ &\quad \times \left[\mathbf{K}_s \mathbf{s} + d \frac{\mathbf{s}}{\|\mathbf{s}\|} \right] \end{aligned} \quad (30)$$

where $\dot{\mathbf{q}}_{eq} = \mathbf{J}^\dagger \dot{\mathbf{x}}_{eq}$ and $\ddot{\mathbf{q}}_{eq} = \mathbf{J}^\dagger (\ddot{\mathbf{x}}_{eq} - \dot{\mathbf{J}} \dot{\mathbf{q}}_{eq})$.

IV. EXPERIMENTAL VALIDATION

A. System Setup

The overview of the hardware configuration is shown in Fig. 2; the Franka Emika Panda seven-DOF manipulator is equipped with a six-axis F/T sensor Nano 25 from ATI Industrial Automation, Inc., which can be used to measure the external force exerted on the end-effector. The Panda manipulator and the ATI Nano 25 F/T sensor are connected to their Control Box and ATI Net Box, respectively. Then, both boxes communicate with the work station PC via Ethernet, which is an Intel Core i9-9900k @ 3.60-GHz desktop computer running with Ubuntu 16.04

LTS (real-time kernel, Linux 4.14.191-rt90) and installed Robot Operating System (ROS) Kinetic. The manipulator is controlled by the libfranka and franka-ros packages under the ROS Kinetic with the sampling period $T_s = 0.001$ s, the environment estimation is also running with 0.001 s, and the dynamic trajectory adaptation loop is running with a sampling period $T_{sr} = 0.005$ s.

In the following experiments, the parameter configuration of the proposed method is the same. The matrix $\mathbf{A}$ in the dynamic compensator (19) is chosen as $\text{diag}[-0.1, \dots, -0.1]$, and $\mathbf{F}_1$ and $\mathbf{F}_2$ in the switching function (20) are chosen as $\text{diag}[0.1, \dots, 0.1]$ and $\text{diag}[10, \dots, 10]$, respectively. Then, the controller gain is chosen as $\mathbf{K}_s = \text{diag}[15, \dots, 15]$. As for the optimization in dynamic trajectory adaptation, it is real-time solved through qpOASES [30] QP library, which can program by C++ and conveniently call in the ROS programming. The weighting $\mathbf{Q}$ and $\mathbf{R}$ are chosen as $\text{diag}[1, \dots, 1]$ and $\text{diag}[\frac{80}{2000^2}, \dots, \frac{80}{2000^2}]$, respectively. Finally, the initial value of $\boldsymbol{\Gamma}$ is chosen as $\boldsymbol{\Gamma}(0) = \text{diag}[1000, 1000]$ in each direction of environment estimation, and the forgetting factor is chosen as $\mu_1 = \mu_2 = 0.9996$.

B. Comparative Experiments: From Motion to Contact

Comparative experiments were carried out with the following algorithms to show the effectiveness of dynamic trajectory adaptation during the transition process even in the case of arbitrary rigid contact. Four control algorithms (C1–C4) were implemented and compared.

The desired motion tracking trajectory is a point-to-point curve, where the max acceleration and velocity are 2 m/s^2 and 0.1 m/s , respectively. All controllers are conducted under two cases with similar contact velocities and environment position uncertainties, i.e., $\mathbf{x}_e - \mathbf{x}_d = 5$ to 10 mm . Case I is a plastic surface, and its deformation is relatively large after being stressed, while Case II is a comparatively stiffer surface.

C1: The proposed approach. The target model parameters are chosen as $\mathbf{M}_d = \text{diag}[1, 1, 1, 0.01, 0.01, 0.01]$, $\mathbf{K}_d = \text{diag}[300, 300, 300, 0.5, 0.5, 0.5]$, $\mathbf{D}_d = 2\sqrt{\mathbf{K}_d \mathbf{M}_d^{-1} \mathbf{M}_d}$, and $\mathbf{K}_f = \mathbf{0}$ in motion tracking, respectively. The initial values of $\mathbf{K}_e$ and $\mathbf{D}_e$ in the dynamic trajectory adaptation are $\mathbf{K}_e = \mathbf{0}$ and $\mathbf{D}_e = \text{diag}[0.001, \dots, 0.001]$ in free space, and changed by RLSE. Also, when the estimated environment stiffness $\mathbf{K}_e[2] \geq 10$ (i.e., environment stiffness in the z -direction), the target model matrices $\mathbf{K}_d$ and $\mathbf{K}_f$ smoothly change to a more compliant target model of $\mathbf{K}_d = \text{diag}[300, 300, 50, 0.5, 0.5, 0.5]$ and $\mathbf{K}_f = \text{diag}[0, 0, -0.5, 0, 0, 0]$, as shown in Fig. 3.

C2: The proposed approach without dynamic trajectory adaptation. The configuration is the same as C1, while not using the replanned trajectories $\mathbf{x}_r(t)$ and $\mathbf{F}_r(t)$.

C3: Impedance control. As in the target model, $\mathbf{K}_d$ and $\mathbf{K}_f$ are chosen as $\mathbf{K}_d = \text{diag}[300, 300, 50, 0.5, 0.5, 0.5]$ and $\mathbf{K}_f = \text{diag}[0, 0, -0.5, 0, 0, 0]$, respectively, both in free-space motion and contact.

C4: Admittance control. The achieved impedance model is the same as C3, while the inner loop is chosen as a motion tracking always; the motion tracking gain is the same as C1 and C2 in free space.

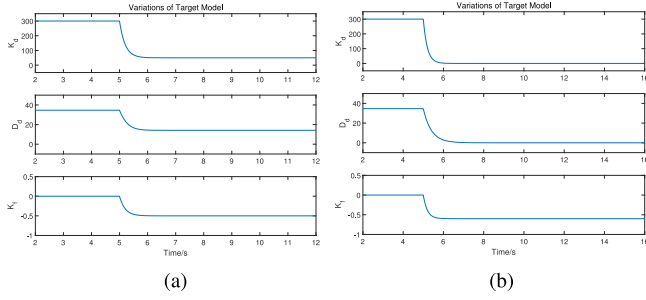


Fig. 3. Variations of the target model. (a) Variation in C1 and C2. (b) Variation in Fig. 8.

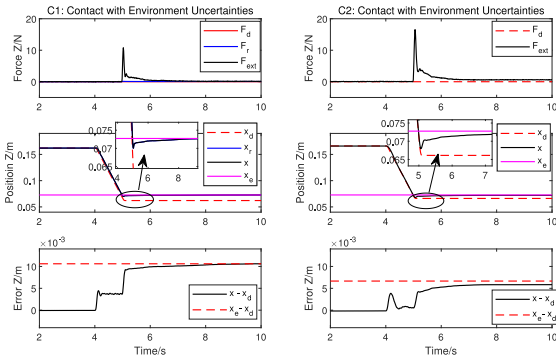


Fig. 4. C1 and C2 under Case I.

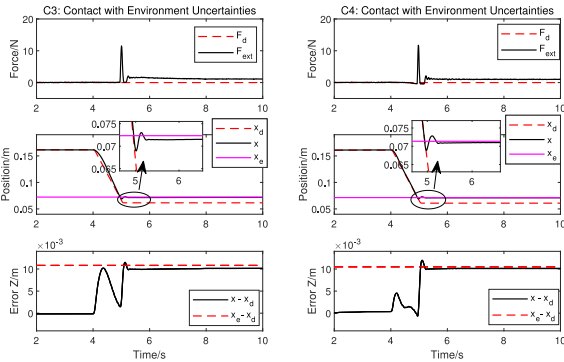


Fig. 5. C3 and C4 under Case I.

As shown in Figs. 4–7, C1–C4 are commanded with the same desired trajectory and contact with two cases (Cases I and II). It should be noted that the line $x - x_d$ represents the motion tracking error only in free space, while the “steady-state” error converges to the environment uncertainties $x_e - x_d$ means that the contact force is small in these figures. From the comparative results, C3 has the most compliant behavior both for Cases I and II since the contact force is small, while the motion tracking performance of C3 in free space is poor compared with other methods. Furthermore, a second contact will occur with C3 when in contact with Case II, as shown in the magnified figure of Fig. 7. Although C4 can improve the motion tracking performance through the inner motion tracking loop, the contact force is larger, especially in Case II, so that a more serious second contact

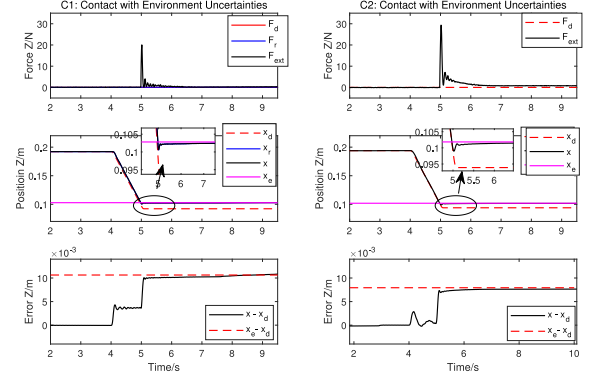


Fig. 6. C1 and C2 under Case II.

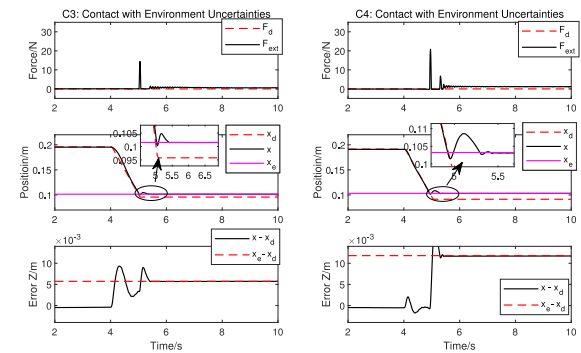


Fig. 7. C3 and C4 under Case II.

will occur than C3. On the contrary, as shown in Figs. 4 and 6, C1 and C2 have a more stable contact establishment than C4 as well as improved free-space motion tracking than C3. However, C2 suffers from a larger contact force than C1 when the contact occurs, especially in Case II. This is because although C1 and C2 both smoothly change the achieved target model of the manipulator from stiff to compliant, as shown in Fig. 3, C1 replans the desired motion trajectory, which can eliminate the mismatch of position and velocity when contacting occurs.

To sum up, the aforementioned comparative experiments show the promising results of the proposed approach. It allows stable and compliant behavior even in the case of arbitrary rigid contact in the unknown environment.

C. Continuous Motion and Contact Force Tracking

Continuous motion and force tracking performances were simultaneously verified in Y and Z directions; an inclined surface was used for contacting in this experiment. The variation of the target model is shown in Fig. 3 from $K_d = \text{diag}[300, 300, 0.5, 0.5, 0.5]$ and $K_f = \mathbf{0}$ to $K_d = \text{diag}[300, 0, 0, 0.5, 0.5, 0.5]$ and $K_f = \text{diag}[0, -0.5, -0.5, 0, 0, 0]$, respectively.

As shown in Fig. 8, the desired motion trajectory of each direction is point-to-point trajectory with maximum velocity 0.05 m/s and maximum acceleration 1 m/s². The desired force trajectory in the Z-direction steps at $t = 7$ s from 0 to 5 N and

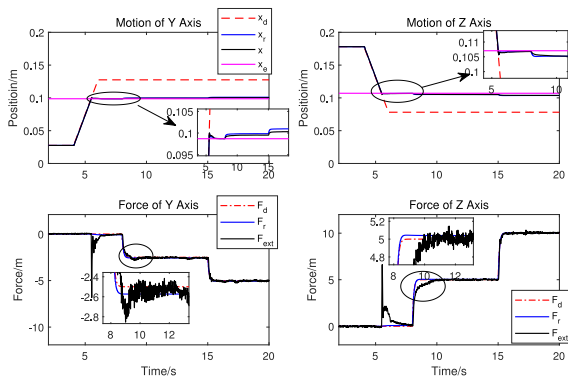


Fig. 8. Continuous motion and force tracking results of Y and Z . The contact point is arbitrarily chosen, and the actual force has noise within ± 0.2 N.

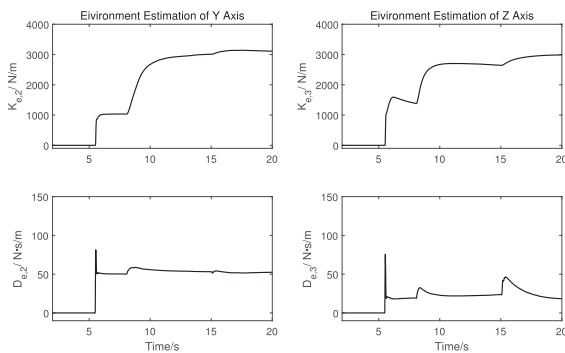


Fig. 9. Environment estimation results.

at $t = 15$ s from 5 to 10 N, and the desired force trajectory in the Y -direction steps at $t = 7$ s from 0 to 2.5 N and at $t = 15$ s from 2.5 to 5 N. The results show that the dynamic trajectory adaptation method replans the desired motion and force trajectories with respect to environment constraints well, where the replanned motion trajectory tracks the desired motion trajectory well before contact and adapts to the environment after contact. After establishing a contact, the target model is smoothly varying to force tracking, so the tracking of the replanned position is not guaranteed. Moreover, with the variation of the target model, the contact force can track the desired force dynamically as well. Meanwhile, the stiffness and damping of environment are estimated and converge, as shown in Fig. 9. To sum up, the manipulator tracks the replanned motion and force trajectories well both in free space and contact.

V. CONCLUSION

A two-loop approach has been developed to achieve the unified motion/force/impedance control for manipulators in unknown contact environments by combining the robust model-reaching control with a dynamic trajectory adaptation method. Specifically, the dynamic trajectory adaptation in the outer loop not only can replan both the motion and force trajectories to conform to the environmental constraints, but also can eliminate the mismatches of the motion and velocity when switching

the control objectives. Meanwhile, the unknown environment constraints are estimated online via the RLSE law, including the stiffness, damping, and equilibrium position. As for the inner loop, the robust model-reaching control is designed to directly realize the compromise between motion and force tracking based on the replanned trajectories of outer loop; stability are guaranteed theoretically. Comparative experiments, continuous motion, and force tracking experiments were conducted to verify the motion and force tracking performance together with the contact compliance during switching. Future works will mainly consist of extending the developed control scheme through reinforcement learning techniques so that the target model can be automatically chosen. Also, further applications in contact-rich tasks of the developed approach will be considered, such as assembly and human-robot interaction.

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